

My Take on Solar Filament Instance Segmentation

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Abstract

This report formalizes the scientific approach implemented in the repository *Solar-Filament-Segmentation-Challenge-2026*. The task is to recover individual solar filaments from full-disk H- α images, with performance evaluated primarily through Panoptic Quality (PQ). The implemented pipeline deliberately separates the problem into two stages: (i) semantic estimation of a filament probability field by an attention-gated U-Net with a ResNet-18 encoder, dilated residual bottleneck, Group Normalization, and a composite pixel/topology loss; and (ii) deterministic conversion of that probability field into filament instances by thresholding, morphological operations, connected components or watershed separation, and instance cleanup.

The statistical emphasis of the pipeline is equally important. Multiple annotation records can correspond to the same physical solar observation, so ordinary random splitting would create direct information leakage. The repository therefore groups folds by physical observation. It also masks the region outside the approximate annotation support, tunes post-processing using the same target metric used by the competition, and reports both image-level and pooled forms of PQ. This report defines every principal object mathematically, explains the motivation and consequences of each modeling choice, and distinguishes verified implementation details from experimental hypotheses described in the notebook. It also identifies current limitations: annotator uncertainty is not modeled explicitly, temporal dependence remains, the solar-limb geometry is approximate, the semantic target discards instance identity during training, and the model scores are not probability-calibrated under the original pixel distribution because of outcome-dependent crop sampling and class-weighted losses.

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1 Scientific problem and scope

1.1 Physical context

A solar filament is a relatively cool and dense structure suspended in the solar atmosphere and seen as an elongated dark structure in H- α full-disk observations. MAGFiLO v1.0, the dataset underlying the challenge, contains manually annotated filament polygons and additional metadata collected from Global Oscillation Network Group (GONG) H- α observations [1]. The released MAGFiLO dataset contains repeated annotations for a non-negligible subset of physical observations, because the annotation protocol allowed independent annotators to label the same image [1].

The competition asks for *instance segmentation*: every predicted filament must be returned as a separate binary mask. The current repository does not directly fit an instance-segmentation network. Instead, it estimates a binary filament field and reconstructs instances in a post-processing stage. This distinction is fundamental.

1.2 Observed data and notation

Let

$$\Omega = \{1, \dots, H\} \times \{1, \dots, W\}, \quad H = W = 2048,$$

denote the pixel lattice. For physical observation i , let

$$X_i : \Omega \rightarrow [0, 1]$$

be the normalized grayscale intensity image. The raw JPEG value is divided by 255, so no additional per-image standardization is applied in the current code.

A physical observation can have several annotation records. Let $\mathcal{A}_i$ be the set of annotators (or, more precisely, annotator-specific COCO records) available for observation i . For annotator $a \in \mathcal{A}_i$, define the observed set of filament instances by

$$\mathcal{G}_{ia} = \{G_{ia1}, \dots, G_{iaK_{ia}}\}, \quad G_{iak} \subseteq \Omega.$$

The semantic union mask used for neural-network training is

$$Y_{ia}(p) = \mathbb{1} \left\{ p \in \bigcup_{k=1}^{K_{ia}} G_{iak} \right\}, \quad p \in \Omega.$$

The unobserved scientific object is better thought of as a latent physical segmentation $\mathcal{G}_i^*$. The sets $\mathcal{G}_{ia}$ are noisy human measurements of that latent object. The current method does *not* fit an explicit annotator-error model; it uses annotator-specific records as training examples and prevents exact physical images from crossing train/validation boundaries.

Statistical unit of analysis

The COCO image record is *not* the independent observational unit. The physical H- α observation is the unit that must be kept intact across data splits. This is why grouped cross-validation is not merely a convenience but a necessary anti-leakage condition.

2 Evaluation target: Panoptic Quality

The challenge changed its primary evaluation criterion from Dice overlap to Panoptic Quality (PQ), while also examining Dice, IoU, fragmentation, over-merging, and efficiency [9]. PQ was introduced by Kirillov et al. [2] to evaluate segmentation and recognition jointly.

2.1 Intersection over Union

For two binary instance masks $A, B \subseteq \Omega$,

$$\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|}.$$

A prediction and a ground-truth instance are eligible to be matched when their IoU is at least 0.5 in the repository’s local evaluator.

Let

$$\hat{\mathcal{G}}_i = \{\hat{G}_{i1}, \dots, \hat{G}_{i\hat{K}_i}\}$$

be predicted instances and

$$\mathcal{G}_i = \{G_{i1}, \dots, G_{iK_i}\}$$

be the ground-truth instances used for evaluation. The code forms the full pairwise IoU matrix and greedily processes candidate pairs in decreasing IoU, enforcing one-to-one matching.

2.2 True positives, false positives, and false negatives

After matching,

$$\text{TP}_i = \{(\hat{G}, G) : \hat{G} \text{ is matched to } G\},$$

while unmatched predictions are false positives and unmatched ground-truth instances are false negatives. Write their cardinalities as

$$TP_i = |\text{TP}_i|, \quad FP_i, \quad FN_i.$$

The segmentation-quality term is

$$\text{SQ}_i = \begin{cases} \frac{1}{TP_i} \sum_{(\hat{G}, G) \in \text{TP}_i} \text{IoU}(\hat{G}, G), & TP_i > 0, \\ 0, & TP_i = 0, \end{cases}$$

and recognition quality is

$$\text{RQ}_i = \frac{TP_i}{TP_i + \frac{1}{2}FP_i + \frac{1}{2}FN_i}.$$

Therefore

$$\text{PQ}_i = \text{SQ}_i \text{RQ}_i = \frac{\sum_{(\hat{G}, G) \in \text{TP}_i} \text{IoU}(\hat{G}, G)}{TP_i + \frac{1}{2}FP_i + \frac{1}{2}FN_i}.$$

The decomposition matters diagnostically:

- low SQ means that matched masks are geometrically inaccurate;
- low RQ means that instances are missed, hallucinated, fragmented, or merged;

- PQ penalizes both simultaneously.

2.3 Two different population summaries

The repository computes both observation-level summaries and a pooled dataset summary. These are different estimands.

The macro, or mean-image, PQ is

$$\overline{\text{PQ}}_{\text{image}} = \frac{1}{n} \sum_{i=1}^n \text{PQ}_i.$$

Every physical observation receives equal weight.

A pooled PQ is instead

$$\text{PQ}_{\text{pooled}} = \frac{\sum_i \sum_{(\hat{G}, G) \in \text{TP}_i} \text{IoU}(\hat{G}, G)}{\sum_i \text{TP}_i + \frac{1}{2} \sum_i \text{FP}_i + \frac{1}{2} \sum_i \text{FN}_i}.$$

Images containing more instances can contribute more strongly to this second quantity. Neither summary is universally “correct”; they answer different questions. Reporting both is scientifically preferable.

Empty images

In the present implementation, an image for which both prediction and ground truth contain zero instances receives $\text{PQ} = 0$, because both SQ and RQ are set to zero in that case. This is a convention of the local implementation and should be checked against any official evaluation code before treating it as definitive.

3 Data splitting and leakage control

3.1 Why ordinary K-fold validation would be invalid

MAGFiLO can contain two or three annotation records of the same underlying solar observation. The image identifier therefore contains annotator information in addition to the physical observation name. If a random split were applied to those COCO identifiers, the same pixel array could appear in both training and validation under two different annotator records. The resulting validation score would estimate performance on a partially memorized image, not generalization to a new solar observation.

The code extracts a grouping key from the physical filename,

$$g(r) = \text{physical observation name of COCO record } r,$$

and applies 5-fold `GroupKFold`. Hence,

$$\{g(r) : r \in \mathcal{D}_{\text{train}}\} \cap \{g(r) : r \in \mathcal{D}_{\text{val}}\} = \emptyset.$$

3.2 Validation deduplication

All annotator-specific records belonging to a training-group observation may remain in the training set. By contrast, the held-out split is deduplicated so that only one annotation record per physical

observation is retained for local validation.

This creates a useful but imperfect design:

- training can expose the network to label variability across annotators;
- validation does not count the same image multiple times;
- however, choosing one annotation record per image does not propagate annotator uncertainty into the reported validation interval.

A more complete statistical model would treat annotation as repeated noisy measurement and either evaluate against a consensus target or integrate over annotator uncertainty.

3.3 Residual dependence

Exact duplicate leakage is removed, but temporal dependence is not. Solar structures evolve over time, and observations close in time can be more similar than observations sampled far apart. Therefore grouped validation by exact observation answers:

“How well does the method generalize to a new image?”

more directly than

“How well does it generalize to a temporally independent solar episode?”.

A date-blocked or active-region-blocked sensitivity analysis is consequently valuable.

4 Annotation-support model near the solar limb

The MAGFiLO annotation instructions exclude filaments near the limb, roughly beyond $\pm 70^\circ$ from the central meridian [1]. Pixels outside the annotation support should therefore not be treated as known negatives.

Let (c_x, c_y) denote the image center and $R = 950$ pixels the fixed approximate disk radius used by the code. Define

$$r(x, y) = \sqrt{(x - c_x)^2 + (y - c_y)^2}.$$

The support first requires

$$r(x, y) \leq R.$$

Under an orthographic projection with solar tilt $B_0 = 0$, horizontal position satisfies

$$x - c_x = R \cos(\phi) \sin(\lambda),$$

where ϕ is latitude and λ longitude relative to the central meridian. At fixed projected ordinate y , the half-width of the disk is

$$h(y) = \sqrt{R^2 - (y - c_y)^2}.$$

The repository therefore approximates the annotated region by

$$V(x, y) = \mathbb{K} \{r(x, y) \leq R, |x - c_x| \leq h(y) \sin(70^\circ)\}.$$

This valid-region mask is used in the loss, in prediction masking, and in instance post-processing.

Missing labels versus negative labels

Outside V , “no annotation” is treated as *missing/undefined supervision*, not as a negative class. This is a principled missing-data decision: penalizing predictions in a region that annotators were instructed not to label would create systematic label bias.

The approximation is nevertheless imperfect because the true solar radius, image centering, and B_0 angle vary between observations. A physically more exact implementation would derive observation-specific disk geometry from metadata.

5 Training-sample construction

5.1 Semantic target

For each annotator-specific record, all filament polygons are rasterized and unioned:

$$Y(p) = \mathbb{1}\{p \text{ belongs to at least one annotated filament}\}.$$

Thus the network is trained for *binary semantic segmentation*. Filament identities are discarded at the neural-loss stage.

This simplifies learning but shifts the burden of instance separation to post-processing. In particular, two nearby filaments whose semantic masks touch are indistinguishable from one connected object unless later morphology separates them.

5.2 Outcome-dependent crop sampling

Full images are 2048×2048 , whereas training uses 768×768 crops. With probability 0.75, when the image contains valid filament pixels, a crop is centered on a random filament pixel. Otherwise the crop origin is sampled uniformly.

Let C denote the random crop and A the event that the crop is filament-centered. Then approximately

$$\Pr(A) = 0.75.$$

The empirical training distribution is therefore not the natural distribution of uniformly sampled image locations. This is intentional: without oversampling, the optimizer would see many mostly-background crops.

The consequence is important. Even if the network output is written as

$$q_\theta(p) = \sigma(z_\theta(p)),$$

it should not automatically be interpreted as an estimate of

$$\Pr(Y(p) = 1 \mid X)$$

under the original pixel population. The sampling design and class-weighted objective alter the effective prior. Threshold calibration against held-out PQ is therefore appropriate.

5.3 Augmentation

The same geometric transformation is applied to the image, target mask, auxiliary array, and valid-region mask. The implemented geometric transformations are generated from horizontal flip,

vertical flip, and rotations by multiples of 90° , which together cover the eight symmetries of the square. Photometric jitter is also applied with probability 0.3:

$$X' = \text{clip}(aX + b, 0, 1)^\gamma,$$

with random a , b , and γ in small intervals.

The geometric augmentation approximately imposes equivariance:

$$f_\theta(TX) \approx Tf_\theta(X).$$

This is useful for morphology-driven segmentation, but it is a modeling assumption rather than a physical identity. Solar filament occurrence is not globally rotationally invariant with respect to heliographic latitude. Since the target task is local segmentation rather than solar-physics inference, the approximation can still be useful, but it should be stated.

6 Neural architecture

6.1 Overview

The implemented model is a U-Net-style encoder–decoder [6]. The default encoder is ResNet-18 [7], instantiated through `timm` with one input channel and no pretrained weights. Encoder feature maps are passed to a decoder through attention-gated skip connections inspired by Attention U-Net [4]. The deepest feature map is first processed by a dilated residual bottleneck.

The default decoder channel sequence is

$$(256, 128, 64, 32, 16).$$

6.2 Why Group Normalization is used

The micro-batch size is one image crop. Batch Normalization would therefore estimate batch moments from a single sample per optimization forward pass. Gradient accumulation increases the *optimizer effective batch size*, but it does not retroactively give Batch Normalization access to a larger forward-pass batch.

The code recursively replaces encoder Batch Normalization layers with Group Normalization [5]. For a feature tensor

$$U \in \mathbb{R}^{C \times H \times W},$$

channels are partitioned into groups. Within each group g , activations are normalized as

$$\tilde{U}_j = \frac{U_j - \mu_g}{\sqrt{\sigma_g^2 + \varepsilon}}, \quad j \in g,$$

followed by learned affine parameters. The key advantage is that μ_g and σ_g^2 do not depend on the number of images in the mini-batch.

6.3 Dilated residual bottleneck

Let F be the deepest encoder tensor. A 1×1 convolution first reduces its channel dimension. The representation is then transformed through 3×3 convolutions with dilation rates

$$d \in \{1, 2, 4\}.$$

A final 1×1 convolution expands the result back to the original channel count, and the branch is added residually:

$$B(F) = \text{ReLU}(F + E(D_4(D_2(D_1(R(F)))))).$$

Dilated convolution increases receptive field without proportionally increasing parameter count or reducing spatial resolution.

The scale parameter of the final normalization layer in the residual branch is initialized to zero. Thus the residual correction begins suppressed. The source-code comment calls this an “exact identity” initialization; strictly, because a final ReLU is applied, the mapping is exactly the identity only where the input is already nonnegative. In practice encoder features are largely activation-processed, so the intended effect is a conservative residual initialization.

6.4 Attention-gated skip connections

For a decoder gating tensor g and encoder skip tensor x , the code computes an additive attention score. Abstractly,

$$\begin{aligned} u &= W_g g + W_x x, \\ \alpha &= \sigma(\psi(\text{ReLU}(u))), \end{aligned}$$

and the gated skip is

$$\tilde{x} = \alpha \odot x.$$

The scalar field $\alpha \in [0, 1]$ is spatially varying and is resized to the skip resolution. It suppresses encoder regions that are less relevant to the decoder’s current context.

This mechanism is plausible for solar-filament segmentation because the target occupies a small portion of the disk and has elongated morphology. It does not, however, constitute an explicit probabilistic attention model; α is simply a learned deterministic gating coefficient.

6.5 Output heads

Two 1×1 output heads are defined:

$$z_{\theta}^{\text{seg}}(p), \quad z_{\theta}^{\text{spine}}(p).$$

The segmentation head is active. The spine head is currently dormant:

- the default spine-loss weight is 0;
- the dataset loader currently returns an all-zero auxiliary distance map;
- a helper for constructing distance-to-spine maps exists but is not used in the training pipeline.

This is consistent with the competition restriction against using other ground-truth metadata for training [9]. The second head should therefore be understood as inactive infrastructure, not part of the current effective estimator.

7 Composite training objective

Let $z_\theta(p)$ be the segmentation logit for pixel p , and

$$q_\theta(p) = \sigma(z_\theta(p)) = \frac{1}{1 + \exp[-z_\theta(p)]}.$$

Only pixels satisfying $V(p) = 1$ contribute to the principal loss.

7.1 Class-weighted binary cross-entropy

For one batch, let

$$N_+ = \sum_p Y(p)V(p), \quad N_- = \sum_p (1 - Y(p))V(p).$$

The positive-class weight is

$$w_+ = \min \left\{ 5, \max \left(1, \frac{N_-}{N_+} \right) \right\}.$$

The masked weighted BCE is

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{\sum_p V(p)} \sum_p V(p) [w_+ Y(p) \log q_\theta(p) + (1 - Y(p)) \log(1 - q_\theta(p))].$$

This term is local and pixelwise. It is especially useful early in optimization but, on its own, can be dominated by class imbalance and says nothing about connectedness of elongated objects.

7.2 Soft Dice loss

For one image, with masked vectors qV and YV ,

$$\text{Dice}_{\text{soft}} = \frac{2 \sum_p q_\theta(p) Y(p) V(p) + \varepsilon}{\sum_p q_\theta(p) V(p) + \sum_p Y(p) V(p) + \varepsilon},$$

and

$$\mathcal{L}_{\text{Dice}} = 1 - \text{Dice}_{\text{soft}}.$$

Dice is an overlap-based objective and is less sensitive than BCE to extreme foreground / background imbalance.

However, Dice can score two masks similarly even when one prediction breaks a long filament into disconnected pieces. That motivates a topology-aware term.

7.3 Soft cIDice

The cIDice family was proposed specifically for tubular and curvilinear structures [3]. Let $S(\cdot)$ denote a differentiable approximation of morphological skeletonization. The code iteratively applies soft erosions and accumulates a soft skeleton.

Define topology precision

$$T_{\text{prec}} = \frac{\sum_p S(q_\theta(p)) Y(p)}{\sum_p S(q_\theta(p))},$$

and topology sensitivity

$$T_{\text{sens}} = \frac{\sum_p S(Y(p)) q_\theta(p)}{\sum_p S(Y(p))}.$$

With numerical smoothing included in the implementation,

$$\text{clDice} = \frac{2T_{\text{prec}}T_{\text{sens}}}{T_{\text{prec}} + T_{\text{sens}}}, \quad \mathcal{L}_{\text{clDice}} = 1 - \text{clDice}.$$

clDice is relevant because solar filaments are thin, elongated, and connectivity-sensitive. It is nevertheless only a surrogate. It is *not* a differentiable version of Panoptic Quality and does not explicitly encode instance identity.

7.4 Total loss

The default effective objective is

$$\mathcal{L}(\theta) = 0.3 \mathcal{L}_{\text{BCE}} + 0.3 \mathcal{L}_{\text{Dice}} + 0.4 \mathcal{L}_{\text{clDice}}$$

because the auxiliary spine term has default weight zero.

This weighting expresses a modeling judgment:

- BCE encourages local classification;
- Dice encourages global foreground overlap;
- clDice receives the largest weight to emphasize connected morphology.

The coefficients are hyperparameters, not likelihood-derived constants.

8 Optimization and regularization

The main defaults are summarized in [table 1](#).

Table 1: Principal configuration values in the reviewed repository.

Component	Parameter	Default
Input	image size	2048×2048
Training crop	crop size	768×768
Crop design	filament-centred probability	0.75
Encoder	architecture	ResNet-18
Pretraining	pretrained weights	false
Normalization	type	Group Normalization
Bottleneck	dilation rates	1, 2, 4
Epochs	training epochs	60
Micro-batch	batch size	1
Gradient accumulation	steps	8
Optimizer	AdamW learning rate	3×10^{-4}
Weight decay	coefficient	10^{-4}
EMA	decay	0.999
Sliding window	size	1024×1024
Window overlap	pixels	256
Default threshold	τ	0.5
Morphological closing	kernel	3×3
Minimum instance area	pixels	50
Validation	grouped folds	5

8.1 AdamW

Parameters are optimized with AdamW [8]. Adam maintains exponential moving estimates of the first and second moments of stochastic gradients and applies coordinate-adaptive steps. AdamW decouples weight decay from the adaptive gradient update, which makes the regularization interpretation cleaner than inserting an L_2 penalty inside Adam’s gradient.

8.2 Gradient accumulation

With micro-batch size 1 and eight accumulation steps, one optimizer update aggregates gradients from up to eight sequential crop losses:

$$g \approx \frac{1}{8} \sum_{j=1}^8 \nabla_{\theta} \mathcal{L}_j.$$

This increases the effective optimization batch size without requiring all eight 768×768 crops in GPU memory simultaneously.

8.3 Cosine learning-rate schedule

The learning rate is annealed over the training horizon by a cosine schedule. Abstractly,

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left[1 + \cos \left(\pi \frac{t}{T} \right) \right].$$

The code uses PyTorch’s `CosineAnnealingLR` with T equal to the configured number of epochs.

8.4 Exponential moving average of parameters

After optimizer steps, an exponential moving average (EMA) is updated:

$$\bar{\theta}_t = \beta \bar{\theta}_{t-1} + (1 - \beta) \theta_t, \quad \beta = 0.999.$$

Validation and saved “best” checkpoints use EMA parameters. The purpose is to smooth high-frequency optimization noise and obtain a more stable inference model.

8.5 Mixed-precision training

Automatic mixed precision is enabled on CUDA. Forward operations can use reduced precision, while gradient scaling reduces underflow risk. The topology loss is explicitly converted to 32-bit floating point for numerical robustness.

9 Full-resolution inference

Training is crop-based, but competition images are full-resolution. Inference therefore uses an overlap-add sliding-window estimator.

Let the inference window size be $w = 1024$, overlap $o = 256$, and stride

$$s = w - o = 768.$$

For each window j , the model produces a probability map $q_j(p)$. If a pixel belongs to $m(p)$ windows, the final field is the arithmetic average

$$\hat{q}(p) = \frac{1}{m(p)} \sum_{j:p \in W_j} q_j(p).$$

Averaging overlapping predictions reduces tile-boundary discontinuities.

9.1 Dihedral test-time augmentation

At inference the code can apply eight square symmetries: identity, flips, transposition, and their compositions. If $\mathcal{T}$ is this transformation set and M is the number of ensemble models, then conceptually

$$\hat{q}(p) = \frac{1}{M|\mathcal{T}|} \sum_{m=1}^M \sum_{T \in \mathcal{T}} T^{-1} [q_{\theta_m}(TX)](p),$$

followed by overlap averaging across windows.

This approximates an equivariant ensemble and reduces variance caused by arbitrary image orientation. It increases computational cost by roughly the number of transformations when all eight are enabled.

10 From semantic probabilities to filament instances

This stage is critical because the leaderboard metric is instance-sensitive even though the network is trained semantically.

10.1 Thresholding and support restriction

For threshold τ ,

$$M_\tau(p) = \mathbb{I}\{\hat{q}(p) > \tau\}V(p).$$

A 3×3 binary closing operation is then applied by default. Closing is dilation followed by erosion and can bridge narrow gaps while removing small concavities.

10.2 Connected-components option

The current default `instance_method` is `components`. Every connected component of M_τ becomes an initial candidate instance.

This has a clear trade-off:

- it does not fragment one elongated connected filament;
- but it merges distinct filaments whenever the thresholded semantic mask connects them.

10.3 Watershed option

The alternative is seeded watershed. Let

$$d(p) = \text{dist}(p, M_\tau^c)$$

be the Euclidean distance transform inside the foreground. Local maxima of d , separated by a minimum distance parameter, define markers. Watershed is then applied to $-d$ inside the foreground mask.

The intention is to split touching objects. For approximately round objects, distance peaks often correspond naturally to instance centers. Solar filaments are instead long and thin; one physical filament can contain many local distance maxima. Consequently watershed can over-segment a single filament into several pieces, which lowers RQ through false-positive and false-negative matching effects.

The notebook explicitly formulates the working hypothesis that connected components may outperform distance-transform watershed for this elongated morphology.

10.4 Per-instance cleanup

For every candidate:

1. small holes up to the configured area threshold are filled;
2. if the candidate contains several disconnected pieces, only the largest connected piece is retained;
3. candidates smaller than `min_area` are dropped;
4. the valid-region mask is reapplied.

The final prediction is therefore a set

$$\hat{\mathcal{G}} = \{\hat{G}_1, \dots, \hat{G}_{\hat{K}}\}.$$

11 Hyperparameter selection and model assessment

11.1 Why post-processing must be tuned on PQ

The network loss and the final metric are not the same function. Threshold τ , minimum area, morphology, and instance-conversion method can substantially change TP, FP, and FN without changing neural weights.

Therefore these parameters should be selected by an empirical risk criterion aligned with the target metric:

$$\hat{\lambda} = \arg \max_{\lambda \in \Lambda} \frac{1}{n_{\text{cal}}} \sum_{i \in \mathcal{I}_{\text{cal}}} \text{PQ}_i(\lambda),$$

where λ collects post-processing parameters.

The generic `tune()` function in `solution.py` searches threshold, watershed minimum-distance, and minimum-area values on the held-out fold.

Current tuning subtlety

The current default instance method is `components`. Under that setting, `min_dist_ws` has no effect. Consequently the generic three-dimensional tuning grid contains repeated equivalent configurations unless the instance method is manually switched to `watershed`. A statistically clean experiment should include the instance method itself as a candidate factor or remove the inactive watershed-distance dimension when `components` are used.

11.2 Notebook-described three-way assessment design

The notebook documents a more rigorous exploratory protocol inside one physical-observation fold:

- 16 fixed observations for model monitoring / checkpoint selection;
- 40 different observations for post-processing calibration;
- 85 remaining observations for paired assessment.

This separation targets three distinct selection stages:

training / checkpoint selection $\rightarrow$ post-processing calibration $\rightarrow$ final local assessment.

It is superior to choosing both network epoch and threshold on the same observations and then reporting that same score as an unbiased estimate.

However, the reviewed public repository contains the main implementation (`solution.py`), the experiment notebook, the dependency list, and a short README. The notebook also refers to `pq_experiment.py`, testing files, reports, and saved run artifacts that are not present in the public repository root at the reviewed revision. Consequently, the three-way protocol is documented in the notebook but cannot be fully independently verified from the available source files alone.

11.3 Paired comparison

Suppose two pipelines A and B are evaluated on the same n observations. Define

$$D_i = \text{PQ}_i^{(B)} - \text{PQ}_i^{(A)}.$$

The natural paired effect estimator is

$$\hat{\Delta} = \frac{1}{n} \sum_{i=1}^n D_i.$$

Pairing removes between-image difficulty from the comparison and is therefore more efficient than comparing two independent sample means.

A nonparametric paired bootstrap resamples observation indices

$$i_1^*, \dots, i_n^*$$

with replacement and recomputes

$$\hat{\Delta}^* = \frac{1}{n} \sum_{b=1}^n D_{i_b^*}.$$

Percentiles of the bootstrap distribution can form a sensitivity interval.

Because temporally adjacent observations may be dependent, a block bootstrap by observation date is a useful robustness analysis. It still does not account for all forms of temporal dependence or training-seed variability.

12 Pipeline summary

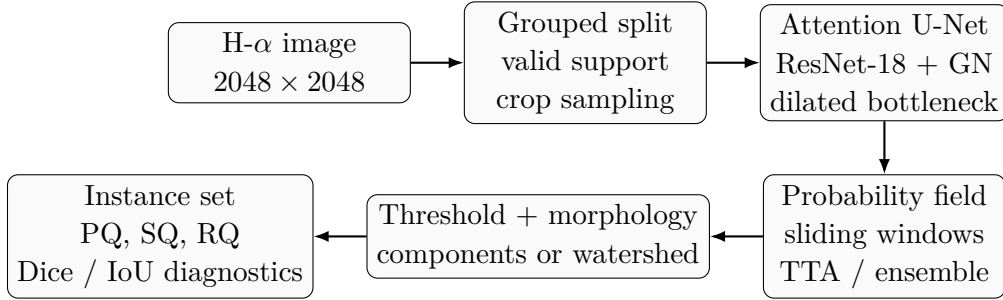


Figure 1: Conceptual structure of the implemented pipeline.

Algorithm 1 Training and evaluation logic

Require: Annotator-specific COCO records and grayscale H- α images

```
1: Extract physical-observation grouping key from filename
2: Build GroupKFold split so no physical observation crosses folds
3: Build approximate valid annotation support  $V$ 
4: for each training epoch do
5:   for each sampled record do
6:     Sample  $768 \times 768$  crop, filament-centred with probability 0.75
7:     Apply geometric and photometric augmentation
8:     Compute logits with attention-gated U-Net
9:     Compute masked BCE + Dice + clDice loss
10:    Accumulate gradients; update AdamW; update EMA parameters
11:   end for
12:   if validation epoch then
13:     Predict full held-out images by sliding windows
14:     Convert probability maps to instances
15:     Compute local PQ and save best EMA checkpoint
16:   end if
17: end for
18: Tune post-processing on held-out/calibration observations using PQ
19: Freeze the selected conversion rule
20: Evaluate on untouched assessment observations where available
```

13 Methodological strengths

From a statistical and scientific perspective, the strongest aspects of the current approach are the following.

Leakage is defined at the physical level. The split respects repeated annotation of the same image rather than pretending COCO rows are independent.

Undefined annotation regions are masked. The model is not penalized for predicting filaments where the labeling protocol deliberately does not provide reliable ground truth.

The objective reflects several error geometries. BCE captures local classification, Dice captures overlap under class imbalance, and clDice adds connectivity sensitivity.

Validation is instance-aware. Checkpoint monitoring and post-processing are tied to PQ instead of optimizing only a pixel-level proxy and hoping it transfers to the competition metric.

Small-batch normalization is handled deliberately. Group Normalization is a defensible choice for a micro-batch size of one.

Inference and training geometry are reconciled. Crop-based training is combined with overlapping full-resolution inference, rather than resizing the solar disk to a much smaller grid and losing thin structures.

14 Limitations and sources of statistical uncertainty

14.1 Inter-annotator variability

MAGFiLO reports substantial but imperfect inter-annotator agreement [1]. The present validation procedure uses one annotation record per physical validation observation. Therefore observed score variability combines:

$$\text{model error} + \text{annotation error} + \text{choice of annotation realization}.$$

A single-score ranking can obscure this decomposition.

14.2 Instance identity is absent from the neural loss

Training on the semantic union

$$Y = \mathbb{K} \left(\bigcup_k G_k \right)$$

removes the distinction between adjacent filament identities. Post-processing must recover that information from geometry alone. This is a structural mismatch between the learning target and the evaluation target.

Possible future alternatives include boundary-aware auxiliary targets, embedding-based instance separation, centerline instance models, or direct instance-segmentation architectures, provided they respect the competition’s ground-truth restrictions.

14.3 clDice is not PQ

clDice encourages connected centerlines, but PQ also requires correct instance counting. A topology-preserving union mask can still have poor recognition quality if separate filaments are merged.

14.4 Probability calibration is not expected

Two mechanisms alter class prevalence during training:

1. positive-centered crop oversampling;
2. batch-adaptive positive weighting in BCE.

Therefore the sigmoid score is best treated as a ranking / segmentation score requiring validation-based thresholding, not as a calibrated posterior probability.

14.5 Approximate observation geometry

The fixed radius $R = 950$, centered disk, and $B_0 = 0$ assumption define a single global support. Real observations can deviate from this geometry.

14.6 Temporal dependence

Grouping exact duplicate observations does not guarantee independence across nearby times. If the scientific claim is generalization across solar conditions rather than leaderboard performance, blocked temporal validation should be added.

14.7 One-fold exploratory experiments

The notebook itself states that its detailed experiment uses one fold and one fine-tuning seed. A single split and seed cannot quantify training randomness or split sensitivity.

15 Recommended statistically rigorous extension

For a final scientific report rather than only a competition submission, the following protocol would strengthen inferential claims.

15.1 Predeclare the estimand

Choose a primary quantity before experimentation, for example

$$\theta = \mathbb{E}_{\text{new physical observation}}[\text{PQ}].$$

State separately whether the target is macro image-level PQ or pooled instance-weighted PQ.

15.2 Use repeated grouped validation

Repeat GroupKFold across several random grouping arrangements where feasible, or use a fixed 5-fold grouped design and repeat training seeds within each fold. Report variation due to:

observation split, optimization seed, post-processing calibration.

15.3 Preserve an untouched assessment set

Do not use the same observations for:

epoch selection, threshold selection, final performance reporting.

The monitor/calibration/assessment separation described in the notebook is a good local template.

15.4 Report uncertainty, not only point estimates

At minimum report:

$$\overline{\text{PQ}}, \quad \text{median}(\text{PQ}_i), \quad Q_{0.05}, \quad Q_{0.95},$$

as the code already supports descriptively, and supplement these with paired bootstrap intervals for method comparisons.

15.5 Analyze the SQ–RQ trade-off

For every ablation, report (SQ, RQ, PQ), not only PQ. For example:

- a lower threshold can increase recall but create FP instances, reducing RQ;
- aggressive watershed can improve separation of merged objects but fragment long filaments, also reducing RQ;
- a topology loss can improve filament continuity while leaving instance separation unchanged.

This decomposition makes architectural claims interpretable.

15.6 Ablation table to include in the final paper

A final report should fill a table of the following form with results obtained under the same assessment protocol.

Table 2: Recommended ablation structure. Numerical entries should be produced from a frozen assessment set; they are intentionally not invented here.

Method	Mean PQ	SQ	RQ	Pixel Dice
BCE only	—	—	—	—
BCE + Dice	—	—	—	—
BCE + Dice + cLDice	—	—	—	—
Components conversion	—	—	—	—
Watershed conversion	—	—	—	—
No TTA	—	—	—	—
8-way TTA	—	—	—	—

16 Code-to-method correspondence

Table 3: Where the scientific components appear in `solution.py`.

Scientific component	Implementation object
Physical-observation grouping	<code>base_name()</code> , <code>get_fold()</code>
COCO polygon rasterization	<code>polygon_to_mask()</code>
Crop sampling and augmentation	<code>FilamentDataset.__getitem__()</code>
Annotation support	<code>build_limb_mask()</code>
Normalization	<code>norm_layer()</code> , <code>replace_batch_norm()</code>
Dilated residual context	<code>DilatedBottleneck</code>
Attention skips	<code>AttentionGate</code>
Segmentation network	<code>FilamentUNet</code>
Dice / cLDice	<code>soft_dice()</code> , <code>soft_cldice()</code>
Total objective	<code>criterion()</code>
EMA	<code>EMA</code>
Training loop	<code>train()</code>
TTA and tiled inference	<code>dihedral_tta_pairs()</code> , <code>predict_full()</code>
Instance conversion	<code>prob_to_instances()</code>
Per-instance cleanup	<code>postprocess_one()</code>
PQ evaluator	<code>match_instances()</code> , <code>panoptic_quality()</code>
Detailed diagnostics	<code>image_metrics()</code> , <code>evaluate_detailed()</code>
Post-processing search	<code>tune()</code>
Submission generation	<code>submit()</code>

17 Interpretation of the overall scientific strategy

The pipeline can be interpreted as a sequence of approximations to a difficult structured prediction problem.

The desired object is a random finite set of filament masks,

$$\mathcal{G}_i^* = \{G_{i1}^*, \dots, G_{iK_i^*}^*\},$$

conditional on an observed H- α image X_i . Directly modeling

$$P(\mathcal{G}_i^* \mid X_i)$$

would require a full instance-level probabilistic model with variable cardinality. The repository instead factorizes the problem pragmatically:

$$X \xrightarrow{f_\theta} \hat{q} \xrightarrow{h_\lambda} \hat{\mathcal{G}},$$

where

$$f_\theta : X \mapsto \hat{q}$$

is a learned semantic score field and

$$h_\lambda : \hat{q} \mapsto \hat{\mathcal{G}}$$

is a deterministic instance-conversion operator controlled by post-processing parameters λ .

Statistically, θ is estimated by minimizing a surrogate empirical risk, while λ is calibrated against the task metric:

$$\begin{aligned} \hat{\theta} &= \arg \min_{\theta} \hat{R}_{\text{surrogate}}(\theta), \\ \hat{\lambda} &= \arg \max_{\lambda} \widehat{\text{PQ}}_{\text{cal}}(h_\lambda(f_{\hat{\theta}}(X))). \end{aligned}$$

This two-stage formulation explains several design decisions that might otherwise seem ad hoc:

- topology-aware neural losses improve the geometry of the continuous score field;
- morphology and connected-component logic supply discrete instance structure;
- PQ calibration aligns the second stage with the final decision criterion;
- a separate assessment sample is needed because λ itself is estimated from data.

The approach is therefore best described not simply as “an Attention U-Net,” but as a **grouped, support-aware, surrogate-risk semantic estimator followed by metric-calibrated structured post-processing**.

18 Reproducibility notes

The reviewed repository exposes deterministic seeds for Python, NumPy, and PyTorch and disables CuDNN benchmarking while requesting deterministic behavior. Checkpoints contain model state, EMA state, optimizer, scheduler, scaler, epoch, configuration, and validation information.

For a publishable experiment, reproducibility should additionally include:

- the exact repository commit hash;

- exact package versions;
- GPU model and CUDA version;
- the precise train / monitor / calibration / assessment observation identifiers;
- all random seeds;
- the selected post-processing configuration;
- wall-clock inference time under a stated hardware environment;
- the official challenge evaluator version or a pinned copy of its logic.

The public notebook references several experiment scripts and saved artifacts that are not part of the currently visible repository root. Those files should be committed, or the notebook should be simplified so that every documented result is reproducible from the public repository.

19 Conclusion

The repository implements a coherent solution to solar-filament instance segmentation under strong class imbalance, thin-object morphology, repeated human annotation, and an instance-sensitive evaluation metric. Its central scientific choices are defensible: physical-observation grouping prevents direct leakage; a valid-region mask respects the annotation protocol; Group Normalization handles the batch-size-one regime; the composite BCE–Dice–cIDice objective combines local accuracy, overlap, and connectivity; overlap-add inference preserves full-resolution detail; and PQ-based post-processing calibration aligns the decision rule with the challenge objective.

The main unresolved issue is not the neural architecture itself but the gap between semantic training and instance-level evaluation. Connected components and watershed represent two opposite error modes: under-separation and over-fragmentation. The statistically strongest next step is therefore to evaluate this trade-off on an untouched paired assessment set, report SQ and RQ alongside PQ, quantify uncertainty across observations and seeds, and avoid interpreting tuned local performance as an unbiased estimate of leaderboard or population performance.

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